BIO 005 HUMAN PHYSIOLOGY · INSTRUCTOR ONLY

Clicker key: Week 9, The Heart as a Pump

Three class sets, picked on the deck's title bar. Concept clickers: 30 s, confidence points. Label-it clickers: 15 s, one tap. Rooms HEART9, HEART9B, HEART9C on the dashboard.

SET A · ROOM HEART9 · KEY: BCADADBCDBAC			ANSWER	
1	CONCEPT	A drug blocks myosin light chain kinase in vascular smooth muscle. What happens to the vessel, and why?	B	It relaxes, because myosin cannot be phosphorylated and cannot cycle
2	LABEL	Name the structure at the pin.	C	Phosphorylated myosin light chain
3	CONCEPT	A toxin destroys the gap junctions in the intercalated discs but leaves the desmosomes intact. What happens to the heartbeat?	A	The cells stay attached but no longer fire together, so the myocardium stops acting as one unit
4	CONCEPT	A patient has a tumor of the adrenal medulla that dumps catecholamines into the blood in bursts. Why do the episodes last many minutes, when a sympathetic nerve response is over in milliseconds?	D	Hormones in blood are cleared slowly by the liver; nerve-released norepinephrine is taken back up at once
5	CONCEPT	A drug blocks HCN (funny current) channels in the SA node. What changes on the pacemaker potential, and what happens to heart rate?	A	The slow depolarization is shallower, threshold takes longer to reach; rate falls
6	LABEL	Name the structure at the pin.	D	Pacemaker potential (slow depolarization)
7	LABEL	Name the structure at the pin.	B	Phase 3: repolarization (ventricle)
8	CONCEPT	On a strip there are 25 small boxes between R waves, and every P is followed by a QRS with a PR of 0.16 s. Rate and rhythm?	C	60 a minute, sinus rhythm
9	LABEL	Name the structure at the pin.	D	T wave
10	CONCEPT	During which phase are all four valves closed and ventricular pressure rising?	B	Isovolumic contraction
11	LABEL	Name the structure at the pin.	A	Isovolumic relaxation
12	CONCEPT	EDV 150 mL, ESV 90 mL, HR 80. Stroke volume, ejection fraction and cardiac output?	C	60 mL, 40%, 4.8 L/min

SET B · ROOM HEART9B · KEY: DBCACDBADBCA			ANSWER	
1	CONCEPT	A tissue's smooth muscle cells all contract together even though only a few of them receive a nerve ending. Which kind of smooth muscle is this, and what makes it possible?	D	Single-unit; gap junctions pass the signal cell to cell
2	CONCEPT	Which pairing of muscle type and calcium source is correct?	B	Cardiac: extracellular trigger Ca^{2+} then SR release
3	LABEL	Name the structure at the pin.	C	Intercalated disc
4	CONCEPT	Why is the adrenal medulla called a modified sympathetic ganglion?	A	Its chromaffin cells are postganglionic neurons without axons, driven by preganglionic cholinergic neurons
5	CONCEPT	A ventricular cell is bathed in a solution with no Ca^{2+}. Which phase of its action potential disappears first, and what happens to the twitch?	C	Phase 2, the plateau; the twitch fails
6	LABEL	Name the structure at the pin.	D	Threshold in the SA node
7	CONCEPT	A strip shows P waves at 80 a minute and wide QRS complexes at 35 a minute with no fixed relationship between them. What is going on?	B	Complete heart block; the ventricles are on a Purkinje escape rhythm

SET B · ROOM HEART9B · KEY: DBCACDBADBCA			ANSWER	
8	LABEL	Name the structure at the pin.	A	Sinoatrial (SA) node
9	LABEL	Name the structure at the pin.	D	ST segment
10	CONCEPT	A murmur is heard between S1 and S2, loudest at the apex. Which valve problem fits?	B	Mitral regurgitation
11	LABEL	Name the structure at the pin.	C	Pulmonic area
12	CONCEPT	A patient's venous return rises suddenly when they lie down. Without any nerve input, what happens to the next stroke volume and why?	A	It rises, because greater EDV stretches the fibers and they contract harder (Frank–Starling)

SET C · ROOM HEART9C · KEY: DBCABADCDBCA			ANSWER	
1	CONCEPT	Smooth muscle in the bladder wall holds tension for an hour using very little ATP. Which feature explains that?	D	The latch state: crossbridges stay attached after the light chain is dephosphorylated
2	LABEL	Name the structure at the pin.	B	Sarcoplasmic reticulum (smooth muscle)
3	CONCEPT	A student says the heart could be tetanized if you stimulated it fast enough, like skeletal muscle. What is the best correction?	C	Wrong; the cardiac refractory period lasts almost as long as the twitch, so a second contraction cannot stack on the first
4	LABEL	Name the structure at the pin.	A	Ryanodine receptor (cardiac SR)
5	CONCEPT	Skeletal muscle arterioles dilate during a fight or flight response even though they have almost no sympathetic nerve supply. Which signal reaches them?	B	Epinephrine from the adrenal medulla acting on β_2 receptors
6	CONCEPT	Which statement correctly contrasts the pacemaker cell with the contractile cell?	A	Pacemaker upstroke is Ca^{2+} and there is no stable resting potential; contractile upstroke is Na^{+} from a stable −90 mV
7	LABEL	Name the structure at the pin.	D	Phase 0: rapid upstroke
8	CONCEPT	Suppose a drug removed the AV node delay. What would go wrong with the pump?	C	The ventricles would contract while the atria are still emptying into them, so ventricular filling drops
9	LABEL	Name the structure at the pin.	D	Purkinje fibers
10	CONCEPT	On a pressure volume loop, afterload rises (the patient's arterial pressure is high). How does the loop change?	B	Taller and narrower: point B is higher and less blood is ejected
11	LABEL	Name the structure at the pin.	C	Point B: aortic valve opens
12	CONCEPT	A β_1 blocker is given. Predict the effect on heart rate and contractility, and name the step that is interrupted.	A	Both fall; norepinephrine can no longer raise cAMP in SA node and contractile cells